

Cloud Computing

UNIT-IV

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Cloud Trends in supporting Ubiquitous Computing

1) Use of Clouds for HPC/HTC

High-performance computing (*HPC*) tasks are characterized as needing large amounts of computing power for short periods of time, whereas *high-throughput computing* (*HTC*) tasks also require large amounts of computing, but for much longer times

Ubiquitous Computing

Ubiquitous computing is a concept in software engineering and computer science where computing is made to appear anytime and everywhere.

In contrast to **desktop computing**, ubiquitous computing can occur using any **device, in any location**, and in any format.

A user interacts with the computer, which can exist in many different forms, **including laptop computers, tablets and terminals** in everyday objects such as a refrigerator or a pair of glasses.

Ubiquitous **cloud computing** refers to the use of Internet resources at any place and any time for any objectives.

A) Herbert Simon on Information Consumption

He says, "What information consumes is rather obvious (**clear**): it consumes the attention of its recipients.

Hence a **wealth of information creates a poverty of attention and a need to allocate that attention efficiently among the overabundance of information sources that might consume it."**

This clearly outlines a **supply and demand model** in the information age.

B)Vannevar Bush on Predicting Memex

He predicted a dream device called Memex that stores all books, records, communications of an individual, and is mechanized so that it may be consulted with exceeding speed and flexibility.

It is an enlarged intimate supplement to his memory.

This Memex is nowadays is iPad, smartphone, cloud,IoT etc.

C) Cloud Service Trends

- Many Companies like Amazon, Enomaly, Google, GoGrid, **Microsoft, NetSuite, Rackspace and Salesforce.com** offer various financial services, communications and media, energy and health care.

D) IBM Cloud Projects

- IBM has gradually changed from a **hardware/software** and system company to a **computing services company**.
- The IBM cloud platforms are mostly built with IBM server clusters **supported by IBM Websphere** and accumulated software assets over the years.
- Both the **z-series** and **p-series** servers at IBM are being upgraded with virtualization capabilities.

E)IBM RC2 Cloud

- The computer and IT infrastructure of **eight IBM Research Centers** are now strongly connected to form a private cloud, called the **Research Compute Cloud(R-C2)**.
- **RC2 is web based**, allowing more than **3,000 IBM researchers worldwide to share computing resources**.
- It is also IBM'S internal testbed(transparent) for promotion of cloud technology.
- The system supports machine virtualization, service life-cycle management and performance monitoring.

F) Cloud System from SGI

- In February 2010, SGI announced the Cyclone, a large-scale on-demand cloud computing service for **high-performance computing (HPC) applications**.
- Both SaaS and IaaS models are offered on the Cyclone.

G) Salesforce.com's Force.com Cloud

- **Salesforce.com** was established in **1999** to provide online solutions to **SaaS**, mainly in **Customer Relationship Management (CRM)** applications.
- Initially, it used third-party cloud platforms to promote its software services.

2) Large scale Private Clouds at NASA and CERN

- In 2010, **two large scale** private clouds were under construction in United States and European Union called, **Nebula is developed by NASA** and is designed for NASA scientists to run **climate models on remote systems provided by NASA**.
- This can save thousands of NASA users from acquiring super computers at their local sites.
- The EU cloud is **built by CERN in Geneva**.
- This is a large private **cloud for distributing data**, applications and computing resources to thousands of scientists around the world.
- CERN deals with large data sets and throughput with a global work force and a finite budget.

a)NASA's Nebula Cloud

- **Nebula** is a **cloud computing pilot** under development at NASA Ames Research Center. It integrates a set of open source components into a seamless, self-service platform, providing high capacity computing, **storage and network connectivity using a virtualized**, scalable approach to achieve cost and energy efficiencies.
- Nebula **enhances NASA'S ability to collaborate** with external scientists and researchers by providing **high-speed data connections and consistent tool sets and open data APIs** used by commercial cloud providers.

A) Nebula Development Status

- Nebula is available to NASA's internal project groups and its research and academic partners.
- **Nebula** is not available for use by private industry or the general public.
- Nebula is also currently acting as a test bed for federal cloud computing technology for the Office of the Management and Budget.

C) CERN Super Cloud

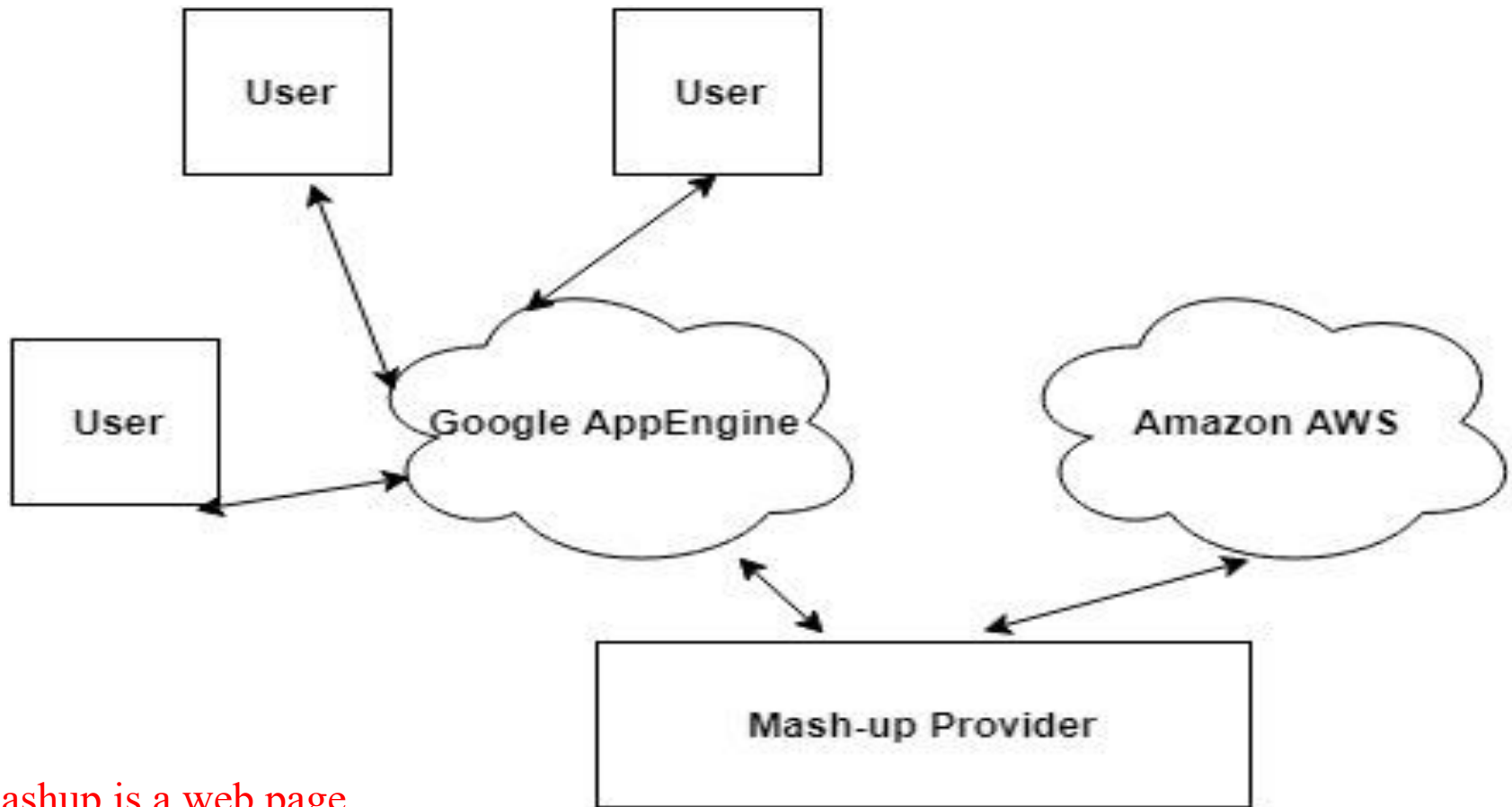
- CERN has long been an influential leader in solving complex IT problems.
- In fact, the World Wide Web owes a lot to CERN for its usefulness in getting some early work done there.
- Now the focus is on cloud computing, CERN scientists plan to smash two particle beams together in an attempt to re-create the big bang event to uncover the enigma of many fundamental physics phenomena.

D) CERN VM Projects

- Scientists working on the **ALICE(Adaptive and scaLable Internet based Computing Engine)** collaboration are conducting heavy ion simulations at CERN.
- They have **been developing and debugging compute** jobs on a collection of internationally **distributed resources, managed** by a scheduler called **AliEn**, to integrate a cloud's dynamically provisioned resources into an existing infrastructure such as the ALICE pool of computers.

3) Cloud Mashups for Agility and Scalability

a) The idea of a Cloud Mashup



a mashup is a web page

- In web application development, a mashup is a web page or application that combines **data, presentations**, or functionality from two or more sources to create a new service.
- The main characteristics of the **mashup** are a combination of **virtualization and aggregation**.
- The **mashup design** leverages Google's web agility with the scaling power of AWS EC2.
- This mashup helps a user to write **agile software** on GAE(**google app Engine**) and use the user inputs to perform parallel computing operations on AWS.

B)Advantages of Cloud Mashup

- *Scalability on the EC2 virtual clusters* Scaling the implementation of **MapReduce** implementation can be scaled in terms of problem size and virtual cluster size.
- *Agility in using GAE interfaces*

A mashup platform is build by combining GAE and AWS, essentially using Google's web interfaces and **Amazon's EC2 computing power**.

We use GAE to create a web page that is **capable of uploading documents to GAE and running MapReduce** on EC2. The mashup then sends the results back to Google users.

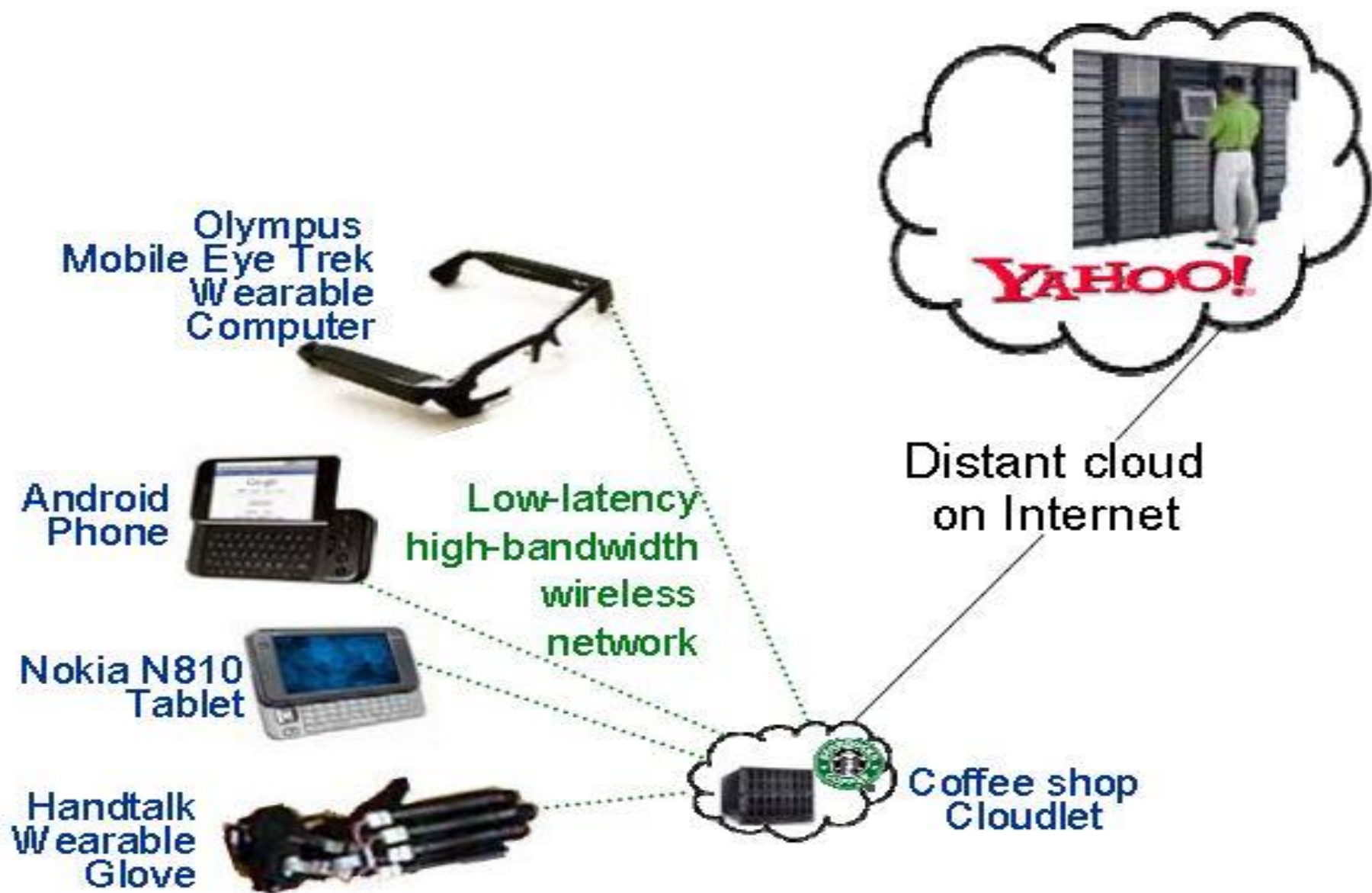
4) Cloudlets for Mobile Cloud Computing

a) The Cloudlet Idea

Mobile handsets face problems of limited CPU power, storage capacity, and network bandwidth on smartphones and tablet computers.

Mobile devices cannot be used to handle large data sets. However, the distant cloud on the Internet faces the WAN-latency problem. Clouds have to solve this collision problem of too many clients logging in to the cloud simultaneously.

To solve this two-sided problem cloudlets are being deployed at convenient sites such as coffee shops and bookstores, similar to access points for WiFi services for connecting to the Internet.



(a) Cloudlet Concept

Differences between Local Cloudlets and Distant Clouds

	Cloudlet	Cloud
State	Only soft state	Soft and hard state
Management	Self-managed; little to no professional attention	Professionally administered
Environment	“Datacenter in a box” at business premises	Machine room with power conditioning and cooling
Ownership	Decentralized by local business	Centralized by Amazon, Yahoo, etc.
Network	LAN latency/bandwidth	Internet latency/bandwidth
Sharing	Few users at a time	100-1000 users at a time

(b) Differences between Cloudlet and Cloud

Performance of Distributed systems and the Cloud

1)Review of Science and Research Clouds

a)Future Grid

Future Grid is a NSF (network file system) project funded as an experimental component of TeraGrid and designed around the concept of Grid5000. FutureGrid is led by Indiana University in partnership with Purdue University ,the San Diego Supercomputer center at the university of California and some other universities.

b)Grid'5000

The Grid'5000 distributed system links nine sites in France and has more than 5000 cores spread over 1500 nodes. It has been used in a variety of computer science research projects including cloud and grid computing and green IT with both software systems and performance goals.

c) Megellan

Megellan is the largest of the **research clouds** and is situated at two U.S. Department of Energy sites: **Argonne and the National Energy Research Scientific Computing Center(NERSC)** in California. The NERSC system has more than 700 nodes and about 6000 cores, while Argonne has more than **500nodes and 4000 cores**

d) Open Cirrus

This is a broad consortium of some 14 partners exploring cloud computing. **The quite powerful test-bed consists of more than 600 nodes at four major sites.** The Open Cirrus testbed is designed to support research into the **design, provisioning and management of services at a global, multidata-center scale.**

e)Open cloud Testbed

- The **Open Cloud Consortium(OCC)** is a member-driven organization that supports the development of standards for cloud computing and frameworks for interoperating between clouds, **develops benchmarks for cloud computing and supports reference implementations for cloud computing.**
- The OCC also manages a testbed for cloud computing, the Open Cloud Testbed and operates cloud computing infrastructure to support scientific research, called **Open Science Data Cloud with an emphasis on data-intensive computing.**

f)Science Clouds

- **Science clouds** is an open cloud federation providing compute cycles in the cloud for scientific communities exploring the use of Nimbus. **The infrastructure federates sites across Europe and United States.**

g)Sky Computing

This project is aligned with Science Clouds and, like the **European Reservoir project**, is designed to federate multiple clouds into a single resource. **Linking separate clouds together** can help scalability, but also improves fault tolerance. The federation also supports the special case of a hybrid **cloud-two clouds, one private and one public.**

h)Venus-C

Venus-C is an European project exploring the applicability of cloud computing in areas such as building structure analysis, 3D architecture rendering, prediction of marine species population, risk prediction for wildfires, metagenomics (study of genetic material recovered from environmental samples), **system biology, and drug development.**

2)Data-Intensive Scalable Computing

a)Big Data Sources

In today's data-driven world, huge databases are generated from science in astronomy, genomics, natural languages and seismic modeling among other applications.

In addition scanned books, newspapers and historical documents are creating a data ocean.

In commerce, corporate sales, stock market transactions, census data, and airline traffic data all demand the handling of large data sets on a daily basis.

b)DISC Systems versus Supercomputers

Table 9.4 DISC Systems versus Supercomputers

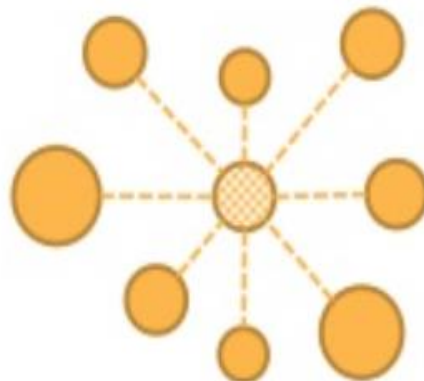
Features	Conventional Supercomputers	Data-Intensive Scalable Computing (DISC)
System architecture	An HPC system by which data is retrieved from remote sites and brought into the system for execution. Heavy data movement overhead.	An HTC data-center cluster which collects and maintains data. Computation collocated with storage for faster access. No data movement overhead.
Programming models	Machine-dependent programs written at a very low level. Use fewer software tools; need specialists to optimize.	Machine-independent application programs on data. Use runtime system controls to optimize execution through load balancing, etc.
System access and use model	Main machine for batch processing when resources are ready. Uses offline visualization at remote site.	Interactive access with priority control and user intervention over large number of users simultaneously.
Reliability issues	Brittle systems with which to recover from most recent checkpoint. Must bring down system for maintenance.	Flexible error detection and recovery. Use redundancy techniques to enter graceful degraded operations in case of failure.

High Performance Computing (HPC) vs. High Throughput Computing (HTC)



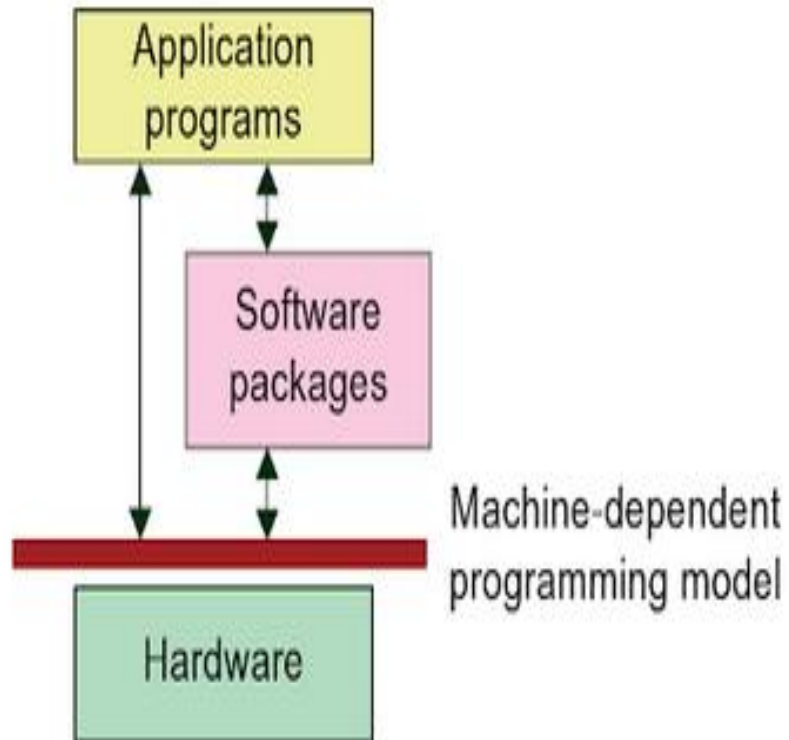
HPC: High performance computing
(cluster computing)

- Tightly clustered
- Latency sensitive

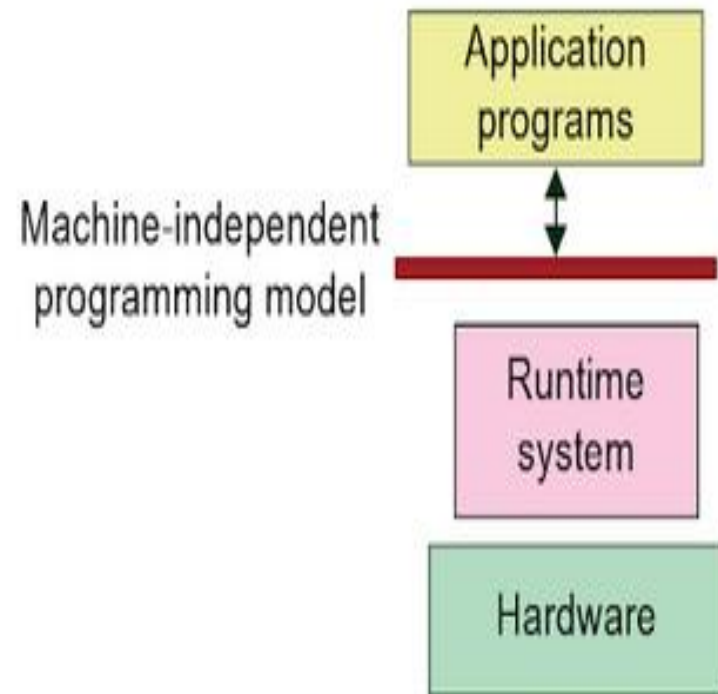


HTC: High throughput computing
(grid computing)

- Less inter-node communication
- More horizontal scalability (pleasingly parallel)



(a) Conventional supercomputer



(b) A DISC system

3)Performance Metrics for HPC/HTC Systems

a)System Throughput and Efficiency

- System **Throughput** of a distributed system or a cloud platform measures the number of jobs that can be done per unit of time.
- The throughput(*deliver to a client within a specified period of time.*) measure is attributed to several key factors that affect the total execution (T_{total}) of all jobs processed in a given time window.
- The overhead can be attributed to five components
 - 1) Infrastructure initiation delay
 - 2) Resource Provisioning delay,
 - 3) Interjob communication delay
 - 4) OS overhead
 - 5) loading application software overhead.

- The OS and software overheads remain constant in running different jobs.
- The other three time factors may vary with problem size and system management policies.
- For simplicity in analysis, we lump all five time factors into a single overhead time kT_0 , where T_0 is their sum averaged over many jobs running in a fixed time interval.
- The constant k is a platform-dependent coefficient that varies with specific system configurations.

- We compute the **total job** execution time with
- $T_{\text{total}} = kT_0 + T_e(n, m)$. Then system throughput is given by $\Pi = n/T_{\text{total}} = n/[kT_0 + T_e(n, m)]$

Where $T_e(n, m)$ be the **effective execution time to complete n independent jobs in a cloud platform**, where m accounts for the number of machine instances in a given system configuration.

The system efficiency(α) is defined by **normalized** throughput as follows:

$$\alpha = \Pi / [n/T_e(n, m)] = T_e(n, m) / [kT_0 + T_e(n, m)]$$

b)Multitasking scalability(B)

- We define multitasking scalability(B) as

$$B=(n/m) \alpha=(n/m) T_e (n,m)/[kT_0 +T_e (n,m)]$$

Where m is the number of machine instances

and n is the number of user jobs being processed and α is system efficiency.

c)System Availability(Y)

- **System availability(Y)** refers to the percentage of time the system is up and running normally.
- This percentage reflects the effects of downtime after unexpected failures and scheduled maintenance for software upgrades.
- System availability is defined by the ratio of **MTTF(mean time to failure)** to the sum of MTTF and MTTR(mean time to repair)

d)Security Index(δ)

- Google has hundreds of data centers and more than half a million servers. Cloud security is attributed to user confidentiality, data integrity, access control, firewalls, IDSes, defense capability against viruses or worm attacks, reputation systems, data lock-in.
- The resource site security index and user access records should be maintained through periodic auditing.

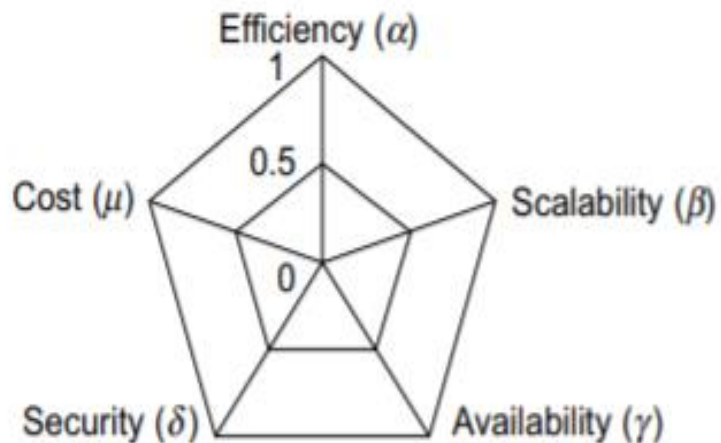
e) Cost Effectiveness(μ)

- The **cost model considers cost** by both the **cloud service provider and the data-center owner**.
- The provider must **pay the data-center owner** for the resources used even if they belong to the same company.
- We denote c as the **hourly charge to the user by the provider** and h as the number of service hours provided to end users.
- The **total income of the provider** from user payment of the service is calculated by $\text{Cost}_p = hc$.
- The total **cost the provider paid to the data center is estimated** by $\text{Cost}_d = h(d/u)$ where **d is the hourly charge** by the data center and u is the utilization rate (often between 0.6 to 0.8) of the data-center resources.
- The **cost-effectiveness(μ)** of the cloud service is a measure for assessing the profit margin by the service provider.

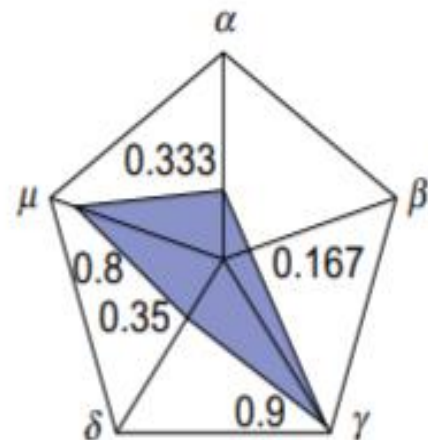
$$\mu = (\text{Cost}_p - \text{Cost}_d) / \text{Cost}_d = (hc - hd/u) / (hd/u) = cu/d - 1$$

4) Quality of Service in Cloud Computing

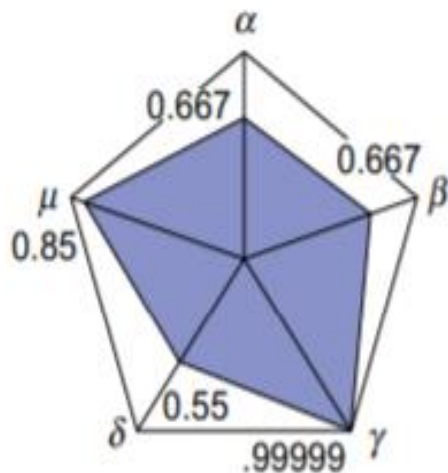
- We calculate the cumulative performance of a cloud platform by a compound metric, called quality of cloud services(QoCS) and denoted by (θ).
- This metric measures the overall quality of the cloud service, regardless of the particular service models applied
- We use a five-dimensional(System efficiency(α), Multitasking scalability(B), System Availability(Y),Security Index(δ), Cost Effectiveness(μ)) Kiviat graph to plot the aggregated cloud performance or to quantify the QoCS graphically.
- $\theta = A_{\text{shaded}} / A_{\text{pentagon}}$ where A_{shaded} and A_{pentagon} are the areas of shaded and outer pentagons as shown in the below figure.



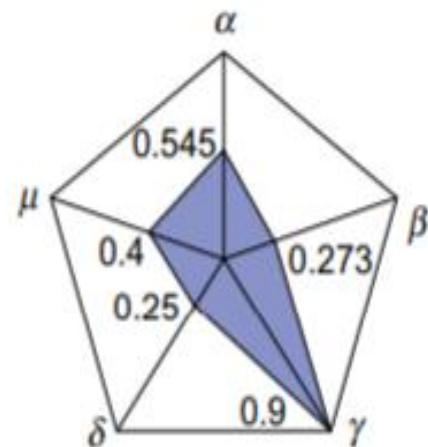
(a) Kiviat graph (template) with five attributes



(b) SaaS with low efficiency, scalability, and security



(c) PaaS under very good running conditions



(d) IaaS under poor running conditions but with HA

ENABLING TECHNOLOGIES FOR THE INTERNET OF THINGS

- In ubiquitous computing, the **Internet of Things(IOT)** provides a network of sensor-or radio connected devices that can be uniquely identified and located in the cyber-physical space.
- The term IOT combines RFID technology with today's IPv6-based **Internet technology**. All things(objects) have IP addresses, which can be uniquely identified.

- The IP-identifiable objects are readable, recognizable, locatable, addressable and controllable via the Internet, aided by **RFID, WiFi, ZigBee, mobile networks and GPS.**

1) The Internet of Things for Ubiquitous Computing

- The IoT is a natural extension of the Internet.
- The foundation of the IOT is radio-frequency identification(RFID).
- This enables the discovery of tagged objects and mobile devices searching by browsing an IP address or searching for a database entry.

a) Ubiquitous Computing

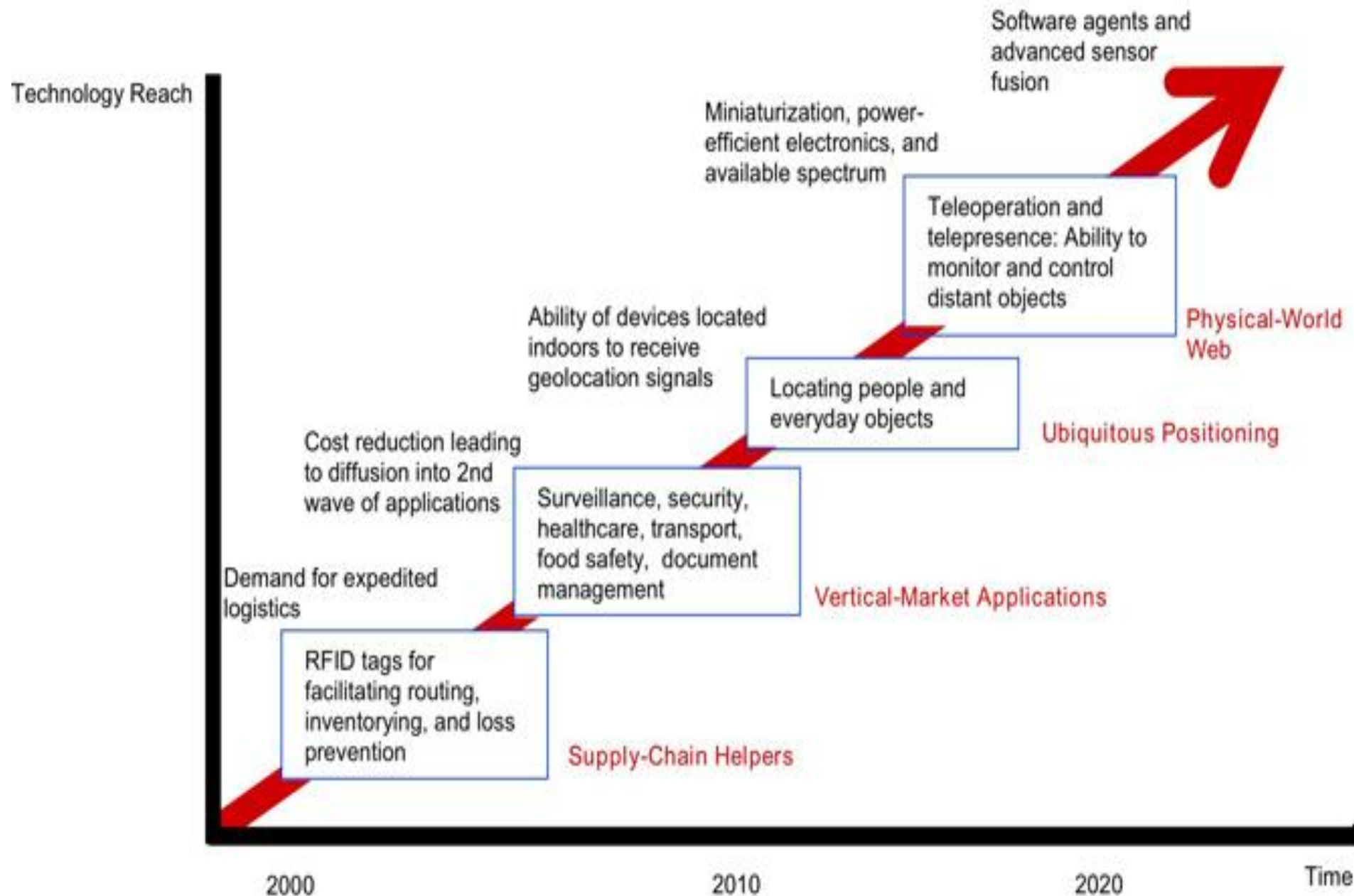
- Ubiquitous computing is a post-desktop model of human-computer interaction in which information processing is integrated into everyday objects and activities.

b) Development of Internet of Things

In 2005, the concept of IoT entered the limelight.

The below figure explains the road map of IoT.

TECHNOLOGY ROADMAP: THE INTERNET OF THINGS



c)Enabling and Synergistic Technologies

Supporting technologies are divided into two categories.

- 1)Enabling Technologies build up the foundations of the IoT. Examples are RFID, sensor networks and GPS are critical.
- 2)Synergistic technologies play supporting roles. Examples are Biometrics, Augmented Reality, Robotics, Machine vision.

Radio-frequency identification(RFID)

d)Architecture of the Internet of Things

- The IoT system is likely to have an event-driven architecture.IoT development is shown with a three–layer architecture.

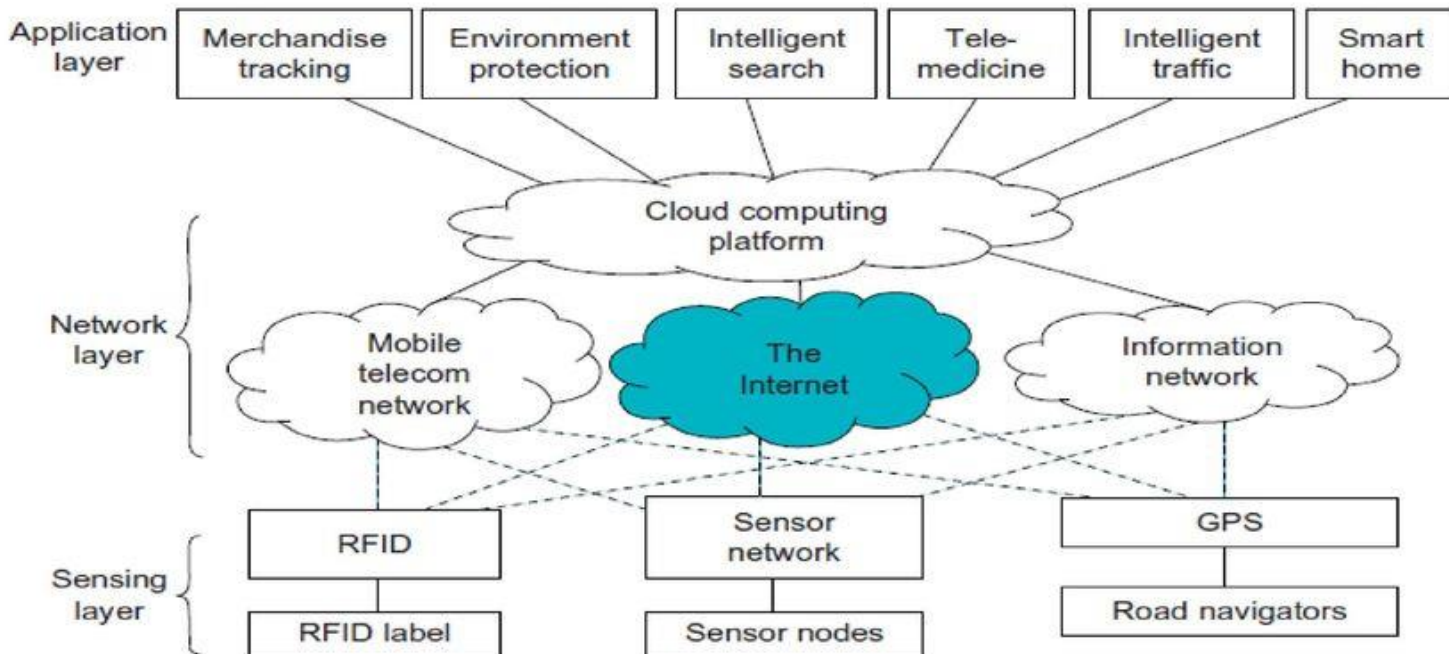


FIGURE 9.15

The architecture of an IoT consisting of sensing devices that are connected to various applications via mobile networks, the Internet, and processing clouds.

Adopted from Ref 1 ch. 9

2)Radio-Frequency Identification(RFID)

- RFID is applied with electronic labels or RFID tags on any objects being monitored or tracked.
- The tagging may be applied to any objects, such as merchandise, tools, smartphones ,computers, animals or people.

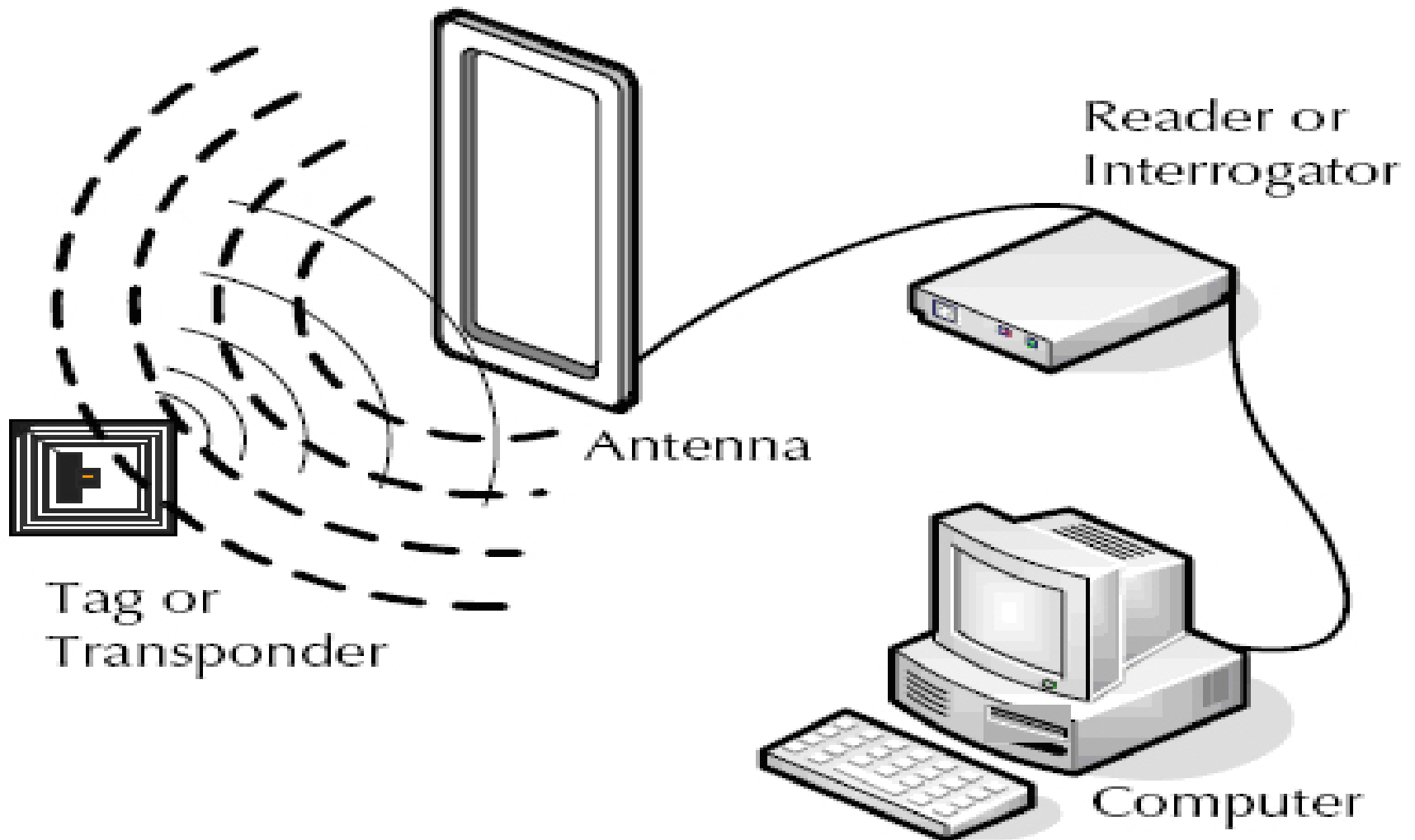
HOW RFID WORKS



a) RFID tags and Device components

- In terms of functionality, there are three major components of RFID hardware:
- **i)RFID tag** : A tiny silicon chip attached to a small antenna
- **ii)Reader Antenna** : Used to radiate the energy and then capture the return signal sent back from the tag. It can be integrated with a handheld reader device or connected to the reader by cable.
- **iii)Reader**: The device station that talks with the tags. A reader may support one or more antennae.

b)How RFID works



- There are two types of RFID tags;
- 1)Active RFID tags
- 2)Passive RFID tags
- **Active** RFID tags contain a battery and transmit signals autonomously.
- **Passive** RFID tags have no battery and require an external source to provoke signal transmission.
- To read the information encoded on a tag, a two-way radio transmitter-receiver called an interrogator or reader emits a signal to the tag using an antenna.
- The tag responds with the information written in its memory bank. The interrogator will then transmit the read results to an RFID computer program.

3) Sensor Networks and ZigBee Technology

- Today's sensor networks are mostly wireless and are known as wireless sensor networks(WSNs).
- A typical WSN consists of spatially distributed autonomous sensors to cooperatively monitor physical or environmental conditions such as temperature, sound, vibration, pressure, motion or pollutants.
- Applications are industrial and civilian application areas, process monitoring and control, machine health monitoring, environment and habitat monitoring, healthcare and home automation and Intelligent traffic control.

a) Wireless Sensor Networks

- The **Sensor network consists of multiple detection** stations called sensor nodes, each of which is small, lightweight and portable.
- **Every sensor node is equipped with a transducer**, microcomputer, transceiver and power source.
- The transducer generates **electrical signals** based on sensed data.
- The microcomputer processes and stores the **sensor output**.
- The transceiver, which can be hardwired or wireless, receives commands from a central computer and **transmits data to that computer**.
- **The power for each sensor node is derived from** the electric utility or from a battery.
- A sensor node may vary in size from that of a shoebox down to a dust grain.

b) Wireless Network Support for Ubiquitous Computing

- Here we discuss four classes of wireless networks for ubiquitous computing. They are
 - i) ZigBee 802.15.4
 - ii) GSM/GPRS/CDMA/1XRTT
 - iii) WIFI 802.11g
 - iv) Bluetooth 802.15.1

Table 9.7 Wireless Networks for Supporting Ubiquitous Computing

Market Name Standard	ZigBee 802.15.4	GSM/GPRS CDMA/1XRTT	WiFi 802.11g	Bluetooth 802.15.1
Application focus	Monitoring and control	Wide area voice and data	Web, e-mail, video	Cable replacement
System Resources	4 KB–32 KB	18 MB+	1 MB+	250 KB+
Battery Life (days)	100–1,000+	1–7	0.5–5	1–7
Network Size	Unlimited (2^{64})	1	32	7
Bandwidth (Kbps)	20–250	64–128+	54,000+	720
Range (meters)	1–100+	1,000+	1–100	1–10+
Success Metrics	Reliability, power, cost	Reach, quality	Speed, flexibility	Cost, convenience

- i) The **ZigBee** networks are mainly used in **low-cost and low-speed monitoring and control applications** such as those used in a wireless home area network (WHAN).
- ii) The **GSM/GPRS/CDMA** networks are cellular mobile networks that can cover a wide area for voice and data telecommunications.
- iii) The **WIFI 802.11g** is used for wirelessly accessing the Internet, reading e-mails and for web search.
- iv) **Bluetooth** is mainly used for short distance wireless connection of computer peripherals.
- Of all above four standards, **ZigBee** has the slowest rate of 20-250kbps.
- However **ZigBee** has high reliability and **low power/cost advantages**.

C) Three generations of Sensor Networks

- i) First Generation(1990's)
- ii) Second Generation(2000's)
- iii) Third Generation(2010's)
- In the first generation, the **sensors were mainly vehicle-placed or air-dropped single sensors.**
- They were quite bulky, like a shoebox and weighed in the kilograms and used in point to point and star topologies.
- In the **second generation, the sensors became smaller, like a pack of playing cards,** and weighed in grams and used in client/server or P2P systems.
- In the **third generation, sensors are as large as dust particles and are used in P2P networks** for embedded and remote applications.

Table 9.8 Three Generations of Wireless Sensor Networks

WSN Features	First Generation (1990s)	Second Generation (2000s)	Third Generation (2010s)
Manufacturers	Custom constructors (e.g., for TRSS)	Crossbow Technology, Inc., Sensoria Corp., Ember Corp.	Dust, Inc., and others
Physical Size	Large shoebox and up	Pack of cards to shoebox	Dust particle
Weight	Kilograms	Grams	Negligible
Node Architecture	Separate sensing, processing, and communication	Integrated sensing, processing, and communication	Integrated sensing, processing, and communications
Topology	Point-to-point, star	Client/server, P2P	P2P
Power Supply	Large batteries; hours, days, and longer	AA batteries; days to weeks	Solar; months to years
Lifetime			
Deployment	Vehicle-placed or air-drop single sensors	Hand-placed	Embedded, sprinkled, left behind

D)ZigBee Network

- ZigBee is a high-level communication protocol using small, low-power, radio-based sensors as specified by the **IEEE 802.15.4 standard**.
- ZigBee devices are applied in WHANs in wireless switches and consumer electronics controlled by short-range radio.
- Zigbee device is simpler to use and less expensive than Bluetooth or WiFi.

There are three different types of ZigBee devices.

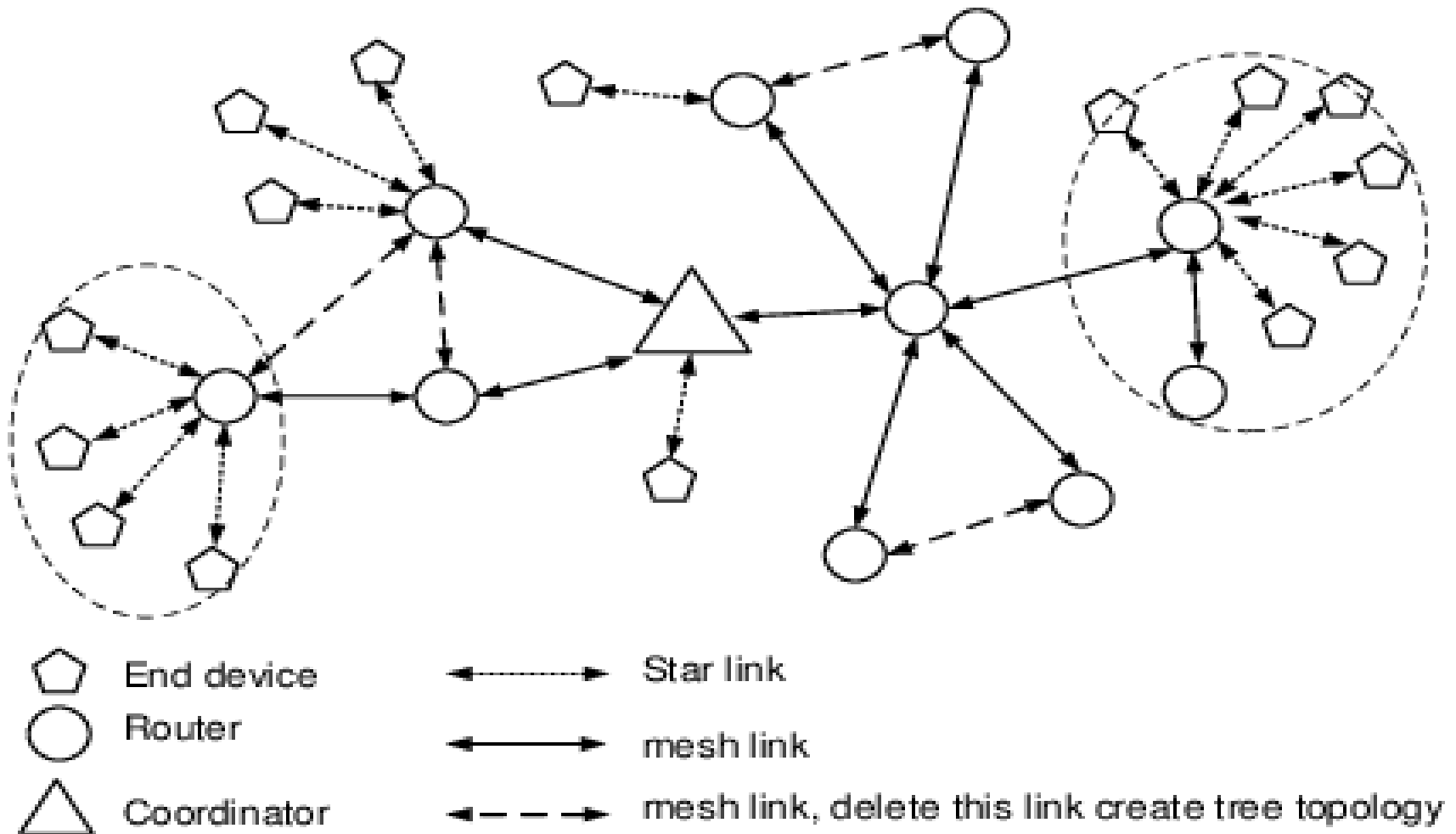
i) ZigBee Coordinator(ZC)

ii) ZigBee Router(ZR)

iii) ZigBee End Device(ZED)

- ZigBee Coordinator is unique to a network which starts the network and acts as **trust center and repository of security keys**.
- ZigBee router acts as a intermediate **router passing on data from end device to end device**.
- ZigBee End device contains just enough functionality to talk to the **parent node**(either the coordinator or the router) and cannot relay data from other devices and **requires least amount of memory**.

Structure of Typical ZigBee network



4)Global Positioning System(GPS)

Location-based service(LBS) helps people and machines find things and determine their physical whereabouts.

GPS consists of three parts.

- i) Space segment
- ii)Control segment
- iii)User Segment

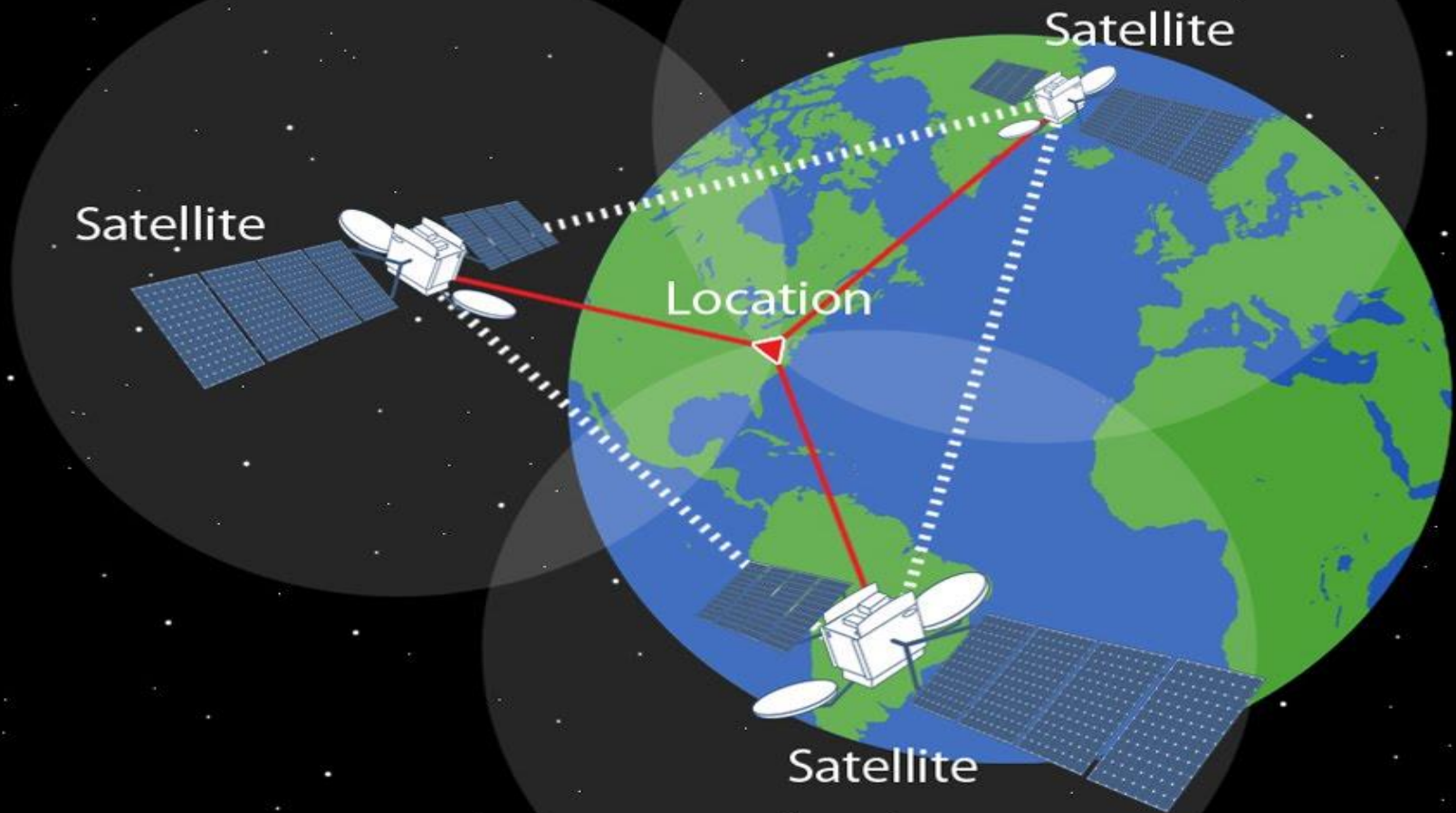
- The **U.S AirForce develops**, maintains and operates the space and control segments. There are 24 satellites deployed around the Earth in fixed orbits at an altitude of approximately 20,200KM.
- GPS satellites broadcast signals from space by which each GPS receiver calculates **its 3D location**(Longitude, Latitude and Altitude) plus the current time.
- The Control Segment is composed of a master control station and a host of dedicated and shared ground antennas and monitor stations.
- The User segment is composed of hundreds of thousands of **U.S. and allied military users of the secure GPS precise positioning services.**

B) Passive versus Active GPSes

- In passive tracking, the GPS is just a receiver, not a transmitter. They lack a transmission capability to send the GPS data **from the vehicle and so are used as loggers and recording devices.**
- Active GPS tracking units incorporate a method to transmit user information from a vehicle.
- Passive GPS tracking devices store GPS location data in their internal memory which can then be downloaded to a computer for viewing at a later time, while active GPS tracking systems send the data at regular intervals to be viewed in real time.

Triangulation

Triangulation



Triangulation

- Knowing the distance from the receiver to a **fixed-position satellite implies that the receiver** is on the surface of a sphere centered at the satellite.
- With four satellites, the receiver location is detected at the intersection of four sphere surfaces.
- The intersection of two satellite spheres is a circle. This circle could be reduced to a single point, if the two spheres merely touch on their surfaces.
- **A circle and a sphere surface intersect at zero ,one or two points .**
- With the **receiver on the surface of Earth**, the receiver needs to choose the point which is closest to the receiver from at most two intersecting points.
- The final receiver location is thus decided by noting the closest point calculated from the two final points to the sphere surface of the fourth satellite.

- A GPS receiver must be locked onto the signal of at least three satellites to calculate a 2D position(longitude, latitude) and track movement
- With four or more satellites in view, the receiver can determine user's 3D position(longitude, latitude and altitude).

Innovative applications of the Internet of Things

- 1) Specific Wireless Sensor Applications
- 2) Retailing and Supply-Chain Management
- 3) Smart Power Grid and Smart Buildings
- 4) Cyber-Physical System(CPS)

1) Specific Wireless Sensor Applications

- **Military networks to detect** and gain as much information as possible about enemy movements, explosions and other phenomena of interest.
- **Sensor networks to detect and monitor environmental changes in plains, forests, oceans and so on.**
- Wireless traffic sensor networks to monitor vehicle traffic on highways or in congested parts of a city
- To detect **and monitor** environmental changes in plains, forests and oceans.

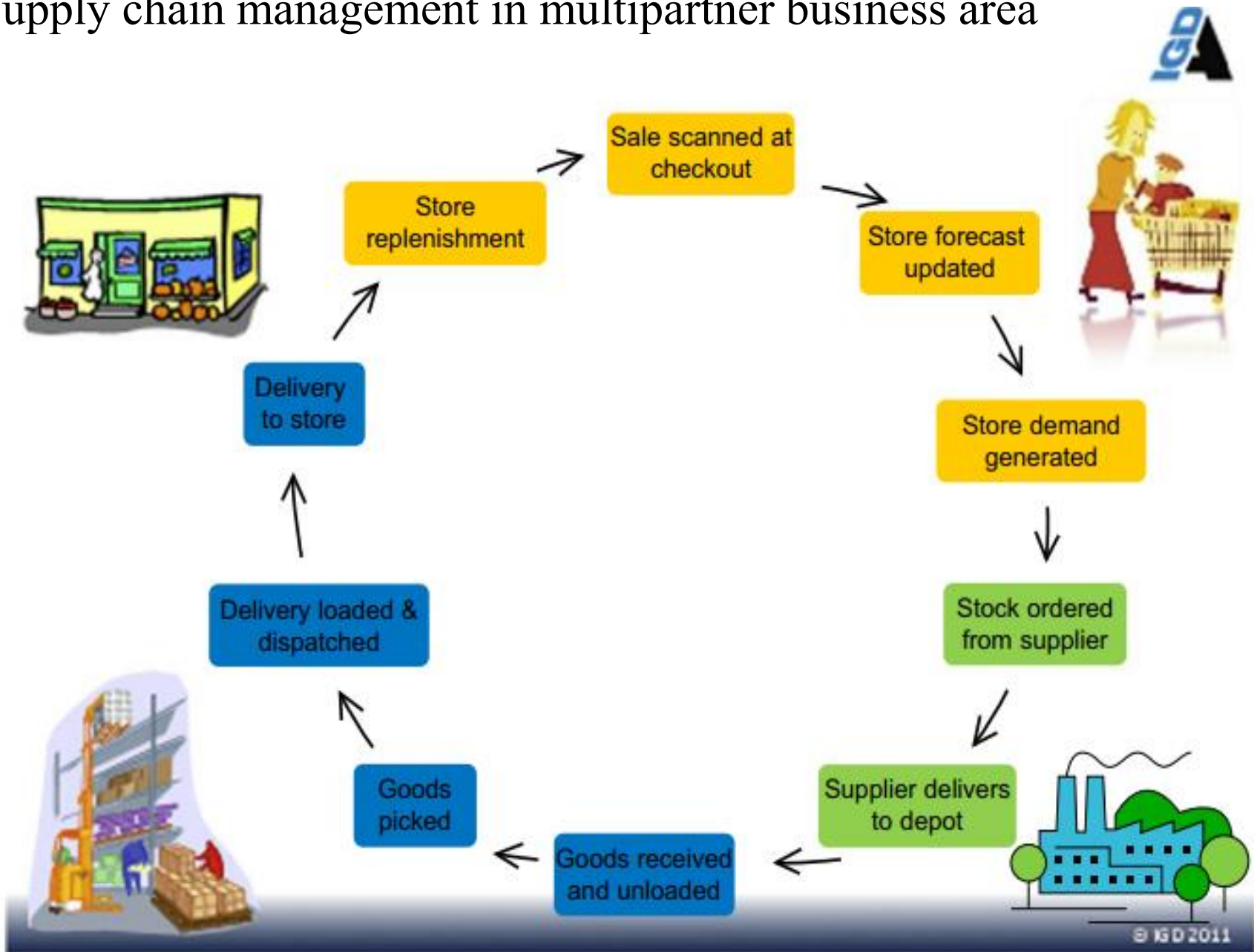
2)Retailing and Supply-Chain Management

- Emergence of **RFID applications** depends strongly on adoption by retailers, logistics organizations and package-delivery companies.
- Retailers may tag individual objects in order to solve a number of problems at once: **accurate inventorying, loss control and ability to support unattended walk-through point** of sale terminals which promise to speed checkout while reducing both shoplifting and labor costs.

2) Retailing and Supply-Chain Management

- The idea is to **manage a whole network of related businesses** or partners involved in product manufacturing, **delivery and services as required by end customers.**

Supply chain management in multipartner business area



3)Smart Power Grid and Smart Buildings

- Various sensors at individual homes (smart thermostats) can collect information that is sent via a network to main stations that can apply complex power management and send control signals back to the grid to save energy.
- Smart buildings may include alarm systems, access controls, indoor climate controls, elevators and so forth.

4)Cyber Physical system(CPS)

- A **cyber-physical system(CPS)** is an embedded system which integrates the computing process with the physical world as an **interactive and intelligent system.**
- Cursor based CPSes can be found in the **automotive, aerospace, health care, robotics, manufacturing, battlefield training and consumer appliance industries.**

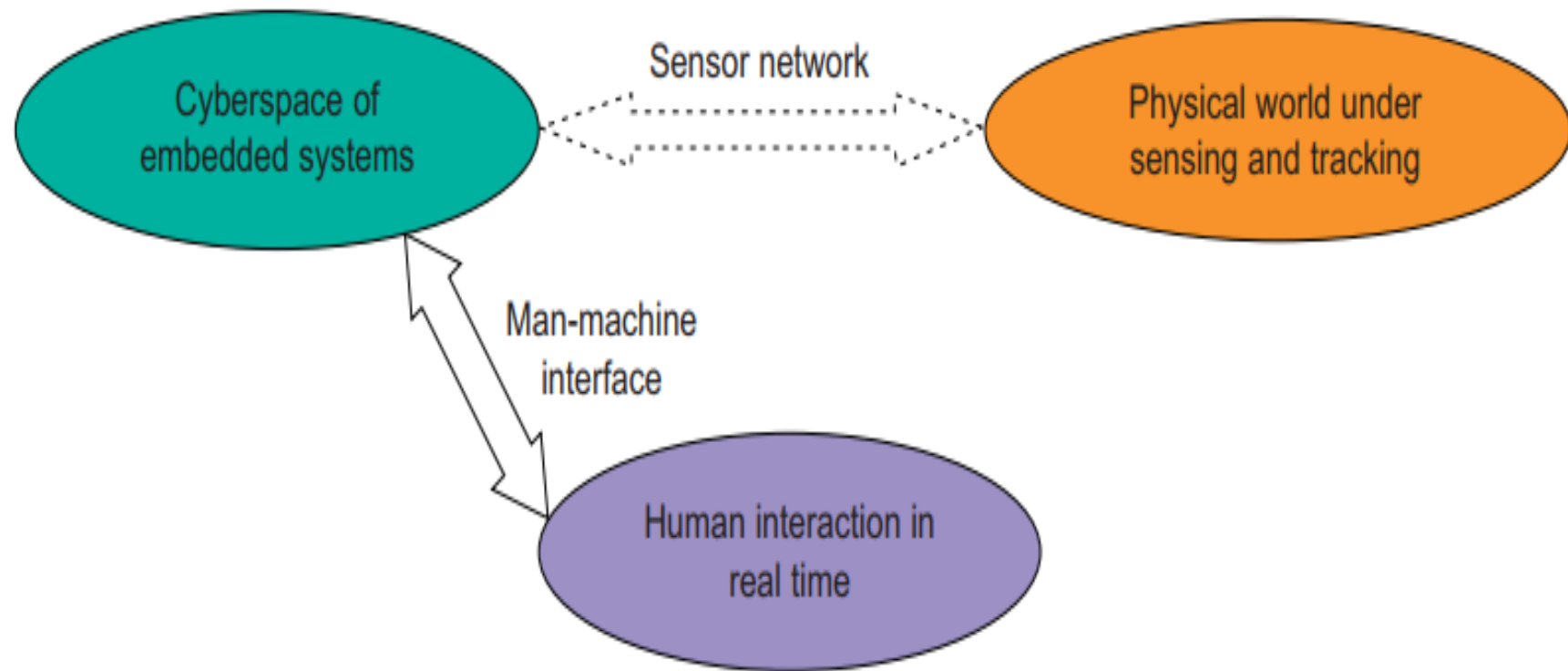


FIGURE 9.23

Three major components working interactively and intelligently in a CPS.

Online Social and Professional Networking

1) Social Network Characteristics

Online social networks(OSNs) are formed with Individual or organizational entities are related, connected, or associated special interests or specific interdependencies.

2) Social Network Services

- a) Personal page or profiles for each user linked by social connections
- b) Social graph traversal along specific social links or networks
- c) Ability to share music, photos and videos with friends or professional groups.
- d) Operation of a community in special niche areas like healthcare, sports and hobbies

3) Social Networks Benefits

- a) **High Return Visit Rate** :Users return to the social network community frequently.This opens up the opportunity for great page impressions and huge advertising inventory.

- **b)User loyalty :** Users connect to their friends and will not abandon them easily.They do not move over to a competing social network.Instead,they show high customer loyalty.
- **c)Virtual Growth:** Members invite their friends to the social network community. This is effective marketing at a low cost and the OSN grows by itself.
- **d)Business model:** With a social network community you can earn revenues through subscriptions to premium content in addition to advertising revenues.
- **4)Social Network Application Domains**
 - a) Business
 - b) Education
 - c) Government
 - d) Medical and Health care
 - e) Online Dating

Thank You